Experiment 1.6

Mapped Course Outcomes-CO4

CO4: Implement VLAN and VLAN Trunking protocols

Aim- Implement VLAN and VLAN Trunking Protocol.

What will you learn:

How VLAN is created using simple connection or creation of VLAN with the help of trunking protocol.

Prerequisites:-

1. A web browser with access to CISCO PACKET TRACER .

Virtual Local Area Network: Virtual LAN (VLAN) is a concept in which we can divide the devices logically on data Link Layer i.e.(Layer2). Generally, Network Layer(layer 3) devices divides broadcast domain and each broadcast domain can be divided by switches using the concept of VLAN.A broadcast domain is a network segment in which if a device broadcast a packet then all the devices in the same network will receive it.However, due to limitations of switches packets don't send outside the broadcast network.To forward out the packets to different VLAN (from one VLAN to another) or broadcast domain, inter VLAN routing is needed. Through VLAN, different small size sub networks are created which are comparatively easy to handle.

Topology

Topology Diagram

Configuration of VLAN:

VLAN can be created by VLAN ID and VLAN name.

Syntax:

#switch1(config)#vlan 2 // Vlan ID

#switch1(config-vlan)#vlan accounts // Vlan Name

Switch(config)#int range fa0/0-2

Switch(config-if) #switchport access Vlan 2

VLAN Trunking Protocol [1]

VLAN Trunking Protocol (VTP) is also a Cisco proprietaryprotocol that propagates the definition of Virtual Local Area Networks (VLAN) on the whole local area network to synchronize the VLAN information in same VTP domain. VTP allows you to add, delete and rename VLANs which is then propagated to other switches in the VTP domain. VTP advertisements can be sent over 802.1Q, and ISL trunk.To do this, VTP carries VLAN information to all the switches in a VTP domain.

Requirements of VTP:

There are some requirements for VTP to communicate VLAN information between switches. These are:

1. The VTP version must be same on the switches user wants to configure
2. VTP domain name must be same on the switches
3. One of the switches must be a server
4. Authentication should match if applied.

Topology for Trunking Protocol

Configuration of Switches.

SwitchA> enable

SwitchA# configure terminal

SwitchA(config)# interface fa 0/24

SwitchA(config-if)#switchport trunk encapsulation dot1q

SwitchA(config-if)# switchport mode trunk

SwitchA(config-if)# end

Conclusion & Discussion.

The trunk is activated. For switch port mode trunk command it is not necessary to use the same command at the other side of the link, without this job it is automatically activated.

Further Readings:

1. https://www.practicalnetworking.net/stand-alone/vlans/